

Neural Networks

CS60050: Machine Learning

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IIT Kharagpur

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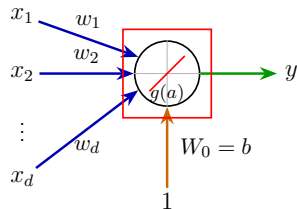
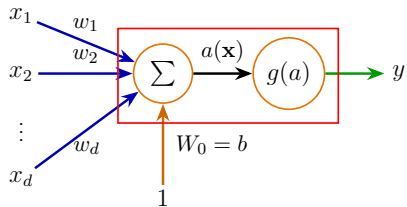
Agenda

- § Introducing the concept artificial neuron networks
- § Understand the forward pass and backward pass

Resources

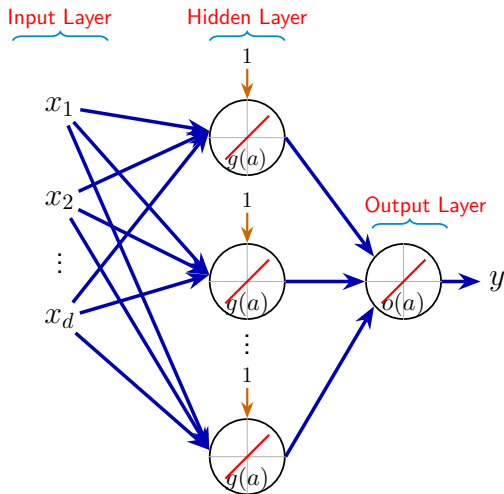
- § Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig.
- § NPTEL Lectures on [Neural Networks and Applications](#) [Lecture 12 & 16] by Prof. Somnath Sengupta, IIT Kharagpur.

Multilayer Neural Network



A bit of notation change

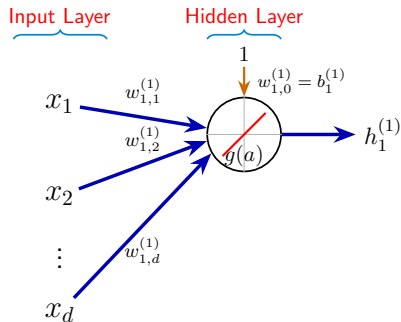
Multilayer Neural Network



Intuition:-

- § **Hidden Layer:** Extracts better representation of the input data.
- § **Output layer:** Does the classification.

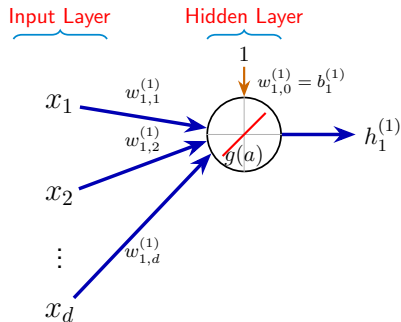
Multilayer Neural Network



$$a_1^{(1)}(\mathbf{x}) = b_1^{(1)} + \sum_{i=1}^d w_{1,i}^{(1)} x_i = \begin{bmatrix} \mathbf{w}_1^{(1)} & b_1^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}$$

$$h_1^{(1)} = g\left(a_1^{(1)}(\mathbf{x})\right)$$

Multilayer Neural Network



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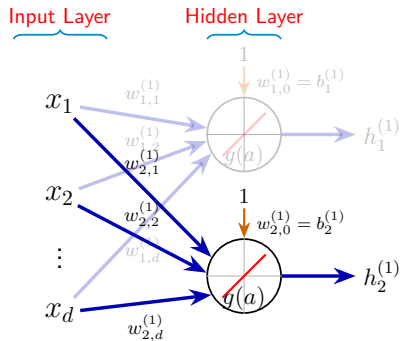
Row vector:

$$\begin{bmatrix} w_{1,1}^{(1)}, w_{1,2}^{(1)}, \dots, w_{1,d}^{(1)} \end{bmatrix}$$

Column vector:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

Multilayer Neural Network



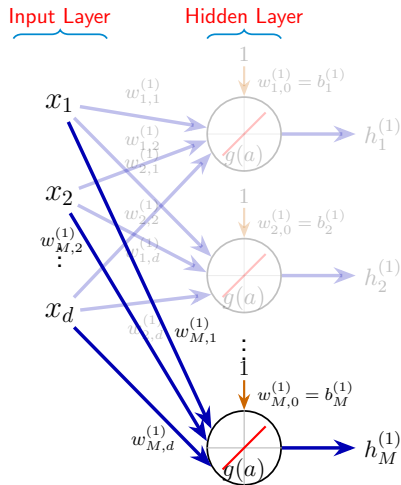
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$$h_1^{(1)} = g\left(a_1^{(1)}(\mathbf{x})\right)$$

$$a_2^{(1)}(\mathbf{x}) = b_2^{(1)} + \sum_{i=1}^d w_{2,i}^{(1)} x_i = \begin{bmatrix} \mathbf{w}_2^{(1)} & b_2^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}$$

$$h_2^{(1)} = g\left(a_2^{(1)}(\mathbf{x})\right)$$

Multilayer Neural Network



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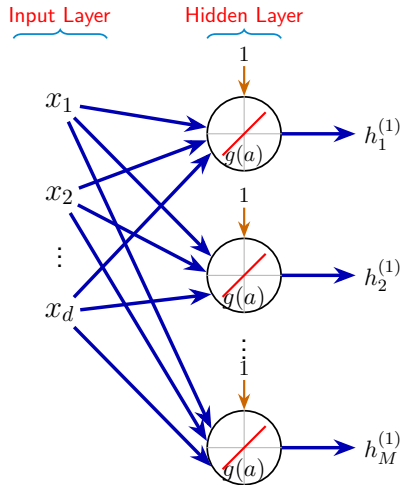
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$$h_2^{(1)} = g(a_2^{(1)}(\mathbf{x}))$$

$$a_M^{(1)}(\mathbf{x}) = b_M^{(1)} + \sum_{i=1}^d w_{M,i}^{(1)} x_i = \begin{bmatrix} \mathbf{w}_M^{(1)} & b_M^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}$$

$$h_M^{(1)} = g(a_M^{(1)}(\mathbf{x}))$$

Multilayer Neural Network



$$\begin{aligned} a_1^{(1)}(\mathbf{x}) &= \begin{bmatrix} \mathbf{w}_1^{(1)} & b_1^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} \\ a_2^{(1)}(\mathbf{x}) &= \begin{bmatrix} \mathbf{w}_2^{(1)} & b_2^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} \\ a_M^{(1)}(\mathbf{x}) &= \begin{bmatrix} \mathbf{w}_M^{(1)} & b_M^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} \end{aligned} \quad \left. \vphantom{\begin{aligned} a_1^{(1)}(\mathbf{x}) \\ a_2^{(1)}(\mathbf{x}) \\ a_M^{(1)}(\mathbf{x}) \end{aligned}} \right\} \begin{bmatrix} a_1^{(1)}(\mathbf{x}) \\ a_2^{(1)}(\mathbf{x}) \\ \vdots \\ a_M^{(1)}(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \mathbf{w}_1^{(1)} & b_1^{(1)} \\ \mathbf{w}_2^{(1)} & b_2^{(1)} \\ \vdots & \vdots \\ \mathbf{w}_M^{(1)} & b_M^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}$$
$$\mathbf{a}^{(1)}(\mathbf{x}) = \begin{bmatrix} \mathbf{W}^{(1)} & \mathbf{b}^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}$$

Column vector Matrix Column vector Column vector

Input Layer



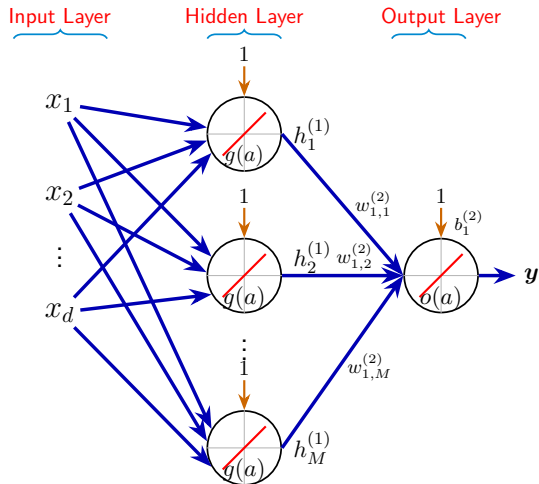
$$\mathbf{a}^{(1)}(\mathbf{x}) = \begin{bmatrix} \mathbf{W}^{(1)} & \mathbf{b}^{(1)} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}$$

Column vector

Hidden Layer Activation

$$\mathbf{h}^{(1)} = g(\mathbf{a}^{(1)})$$

Multilayer Neural Network



Output Layer Pre-activation:

$$a_1^{(2)} \left(h^{(1)} \right) = \begin{bmatrix} w_1^{(2)} & b_1^{(2)} \end{bmatrix} \begin{bmatrix} h^{(1)} \\ 1 \end{bmatrix}$$

Output Layer Activation:

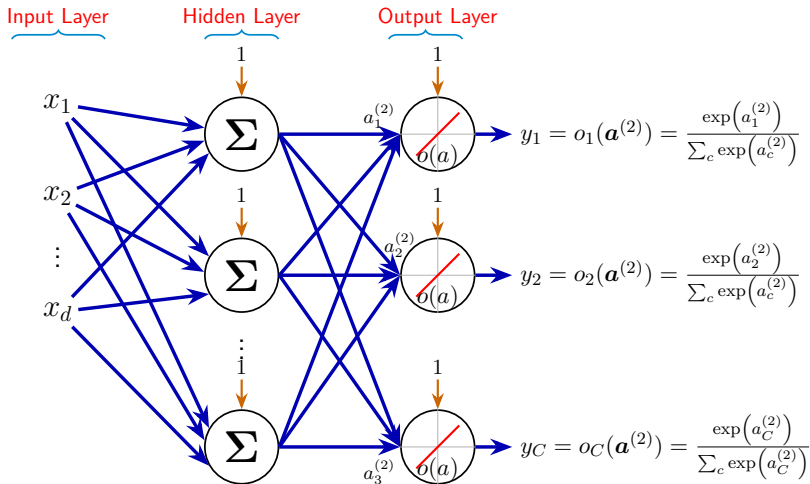
$$y = o \left(a_1^{(2)} \left(h^{(1)} \right) \right)$$

o can have many options:

- § sigmoid
- § tanh
- § ReLU etc.

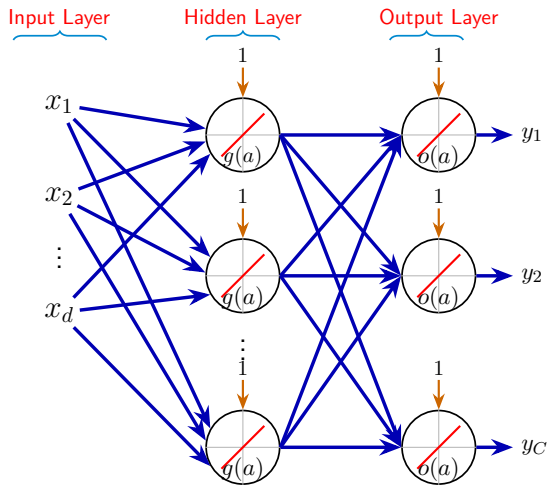
One special is **softmax** function.

Multilayer Neural Network

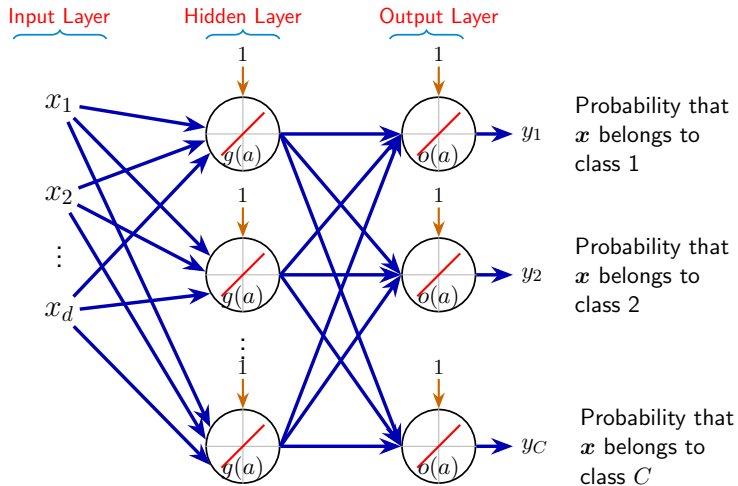


Softmax function:
Exponentiate each component of output activation vector and elementwise divide by the sum of the exponentials.

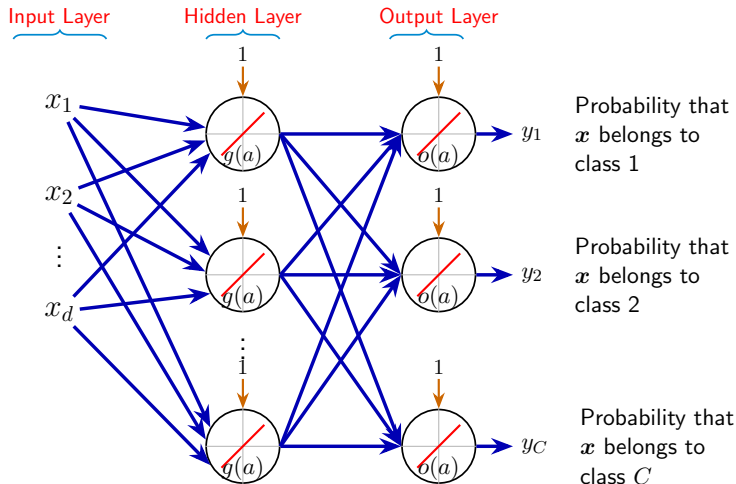
Training a Neural Network - Loss Function



Training a Neural Network - Loss Function

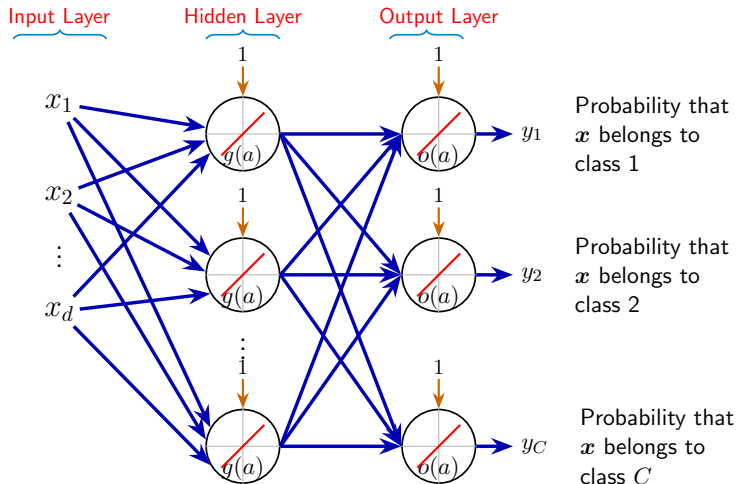


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So we can aim to maximize the probability corresponding to the correct class for any example x

Training a Neural Network - Loss Function

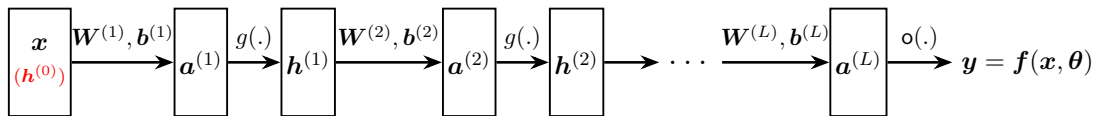


So we can aim to maximize the probability corresponding to the correct class for any example x

$$\begin{aligned} & \max \mathbf{y}_c \\ & \equiv \max (\log \mathbf{y}_c) \\ & \equiv \min (-\log \mathbf{y}_c) \end{aligned}$$

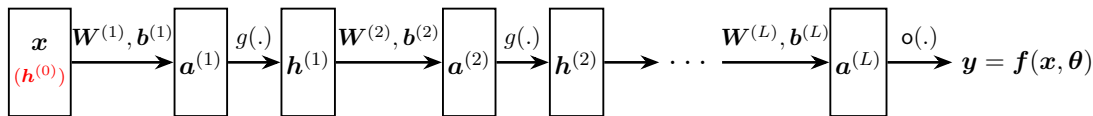
Can be equivalently expressed as $-\sum_i \mathbb{I}(i = c) \log(\mathbf{y}_i)$ known as cross-entropy loss.

Forward Pass in a Nutshell



θ is the collection of all learnable parameters i.e., all W and b

Forward Pass in a Nutshell



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§ Hidden layer pre-activation:

For $l = 1, \dots, L$; $a^{(l)} = W^{(l)}h^{(l-1)} + b^{(l)}$

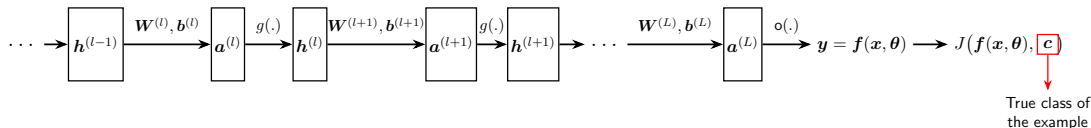
§ Hidden layer activation:

For $l = 1, \dots, L - 1$; $h^{(l)} = g(a^{(l)})$

§ Output layer activation:

For $l = L$; $h^{(L)} = o(a^{(L)}) = f(x, \theta)$

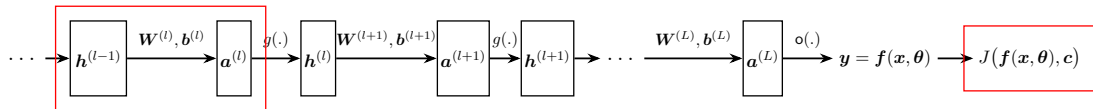
Backpropagation



For gradient descent we need to compute $\frac{\partial J}{\partial W_{i,j}^{(l)}}$ and $\frac{\partial J}{\partial b_i^{(l)}} \quad \forall i, j, l$

Backpropagation consists of intelligent use of the chain rule of differentiation for computation of these gradients.

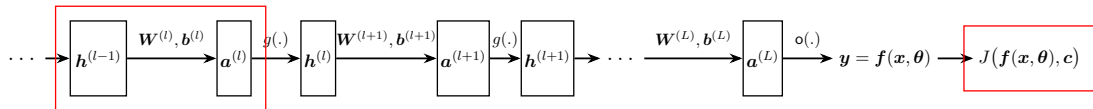
Backpropagation



$W_{i,j}^{(l)}$ influences J through $a_i^{(l)}$. That means J is a function of $a_i^{(l)}$ and $a_i^{(l)}$, in turn, is a function of $W_{i,j}^{(l)}$

$$\frac{\partial J}{\partial W_{i,j}^{(l)}} = \frac{\partial J}{\partial a_i^{(l)}} \frac{\partial a_i^{(l)}}{\partial W_{i,j}^{(l)}}$$

Backpropagation

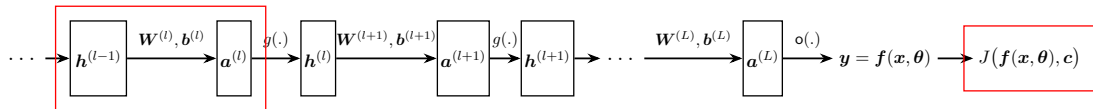


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$$a_i^{(l)} = \sum_j W_{i,j}^{(l)} h_j^{(l-1)} + b_i^{(l)} = W_{i,1}^{(l)} h_1^{(l-1)} + W_{i,2}^{(l)} h_2^{(l-1)} + \dots + W_{i,j}^{(l)} h_j^{(l-1)} + \dots + b_i^{(l)}$$

Backpropagation



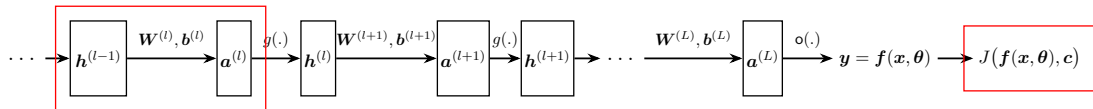
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Backpropagation



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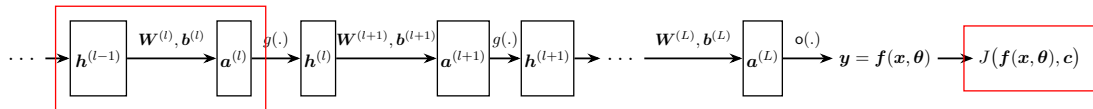
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And similarly
$$\frac{\partial J}{\partial b_i^{(l)}} = \frac{\partial J}{\partial a_i^{(l)}} \frac{\partial a_i^{(l)}}{\partial b_i^{(l)}} = \frac{\partial J}{\partial a_i^{(l)}} \quad \dots (2)$$

Backpropagation



$W_{i,j}^{(l)}$ influences J through $a_i^{(l)}$. That means J is a function of $a_i^{(l)}$ and $a_i^{(l)}$, in turn, is a function of $W_{i,j}^{(l)}$

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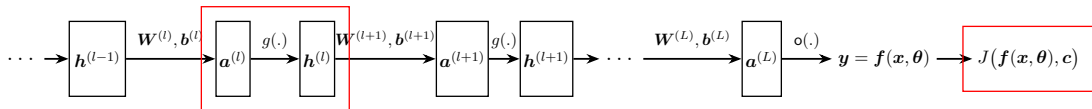
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We know $h_j^{(l-1)}$ from forward pass,

so we need $\frac{\partial J}{\partial a_i^{(l)}}$

Backpropagation

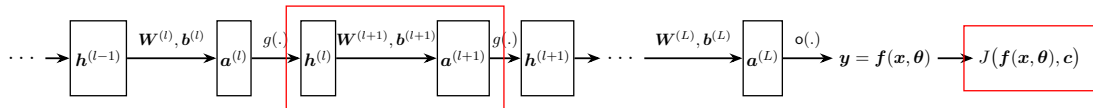


$a_i^{(l)}$ influences J through $h_i^{(l)}$. That means J is a function of $h_i^{(l)}$ and $h_i^{(l)}$, in turn, is a function of $a_i^{(l)}$

$$\frac{\partial J}{\partial a_i^{(l)}} = \frac{\partial J}{\partial h_i^{(l)}} \frac{\partial h_i^{(l)}}{\partial a_i^{(l)}} = \frac{\partial J}{\partial h_i^{(l)}} g' \left(a_i^{(l)} \right) \quad \dots (3)$$

We know $g' \left(a_i^{(l)} \right)$ from forward pass, so we need $\frac{\partial J}{\partial h_i^{(l)}}$

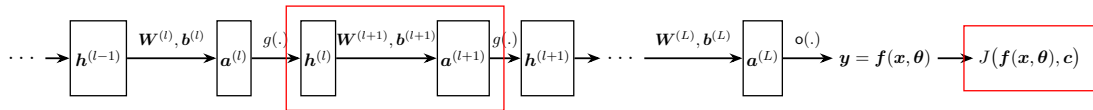
Backpropagation



$h_i^{(l)}$ influences J through $a_{1:M_{l+1}}^{(l+1)}$. That means J is a function of $a_{1:M_{l+1}}^{(l+1)}$ and each of $a_{1:M_{l+1}}^{(l+1)}$, in turn, is a function of $h_i^{(l)}$.

Number of neurons in hidden layer $(l + 1)$

Backpropagation



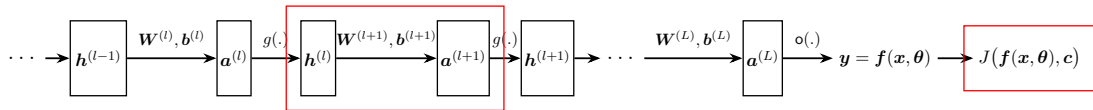
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Number of neurons in hidden layer $(l + 1)$

$$\frac{\partial J}{\partial h_i^{(l)}} = \sum_{m=1}^{M_{l+1}} \frac{\partial J}{\partial a_m^{(l+1)}} \frac{\partial a_m^{(l+1)}}{\partial h_i^{(l)}}$$

Remember If, $u = f(x, y)$, where $x = \phi(t)$, $y = \psi(t)$, then $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial t}$

Backpropagation



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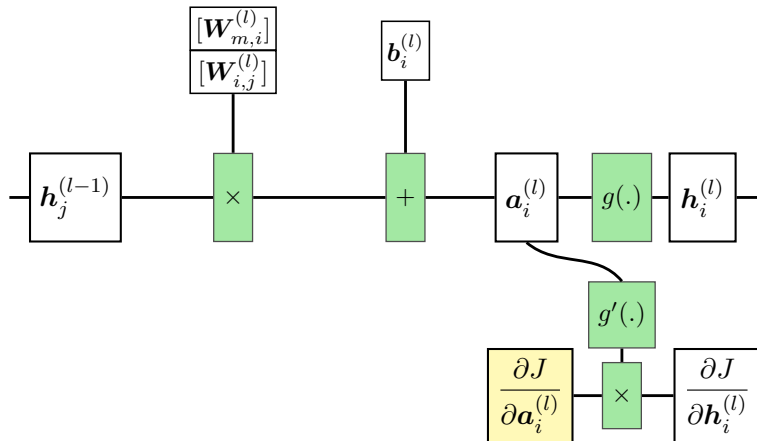
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$$a_m^{(l+1)} = \sum_j W_{m,j}^{(l+1)} h_j^{(l)} + b_m^{(l+1)} = W_{m,1}^{(l+1)} h_1^{(l)} + W_{m,2}^{(l+1)} h_2^{(l)} + \dots + W_{m,i}^{(l+1)} h_i^{(l)} + \dots + b_m^{(l+1)}$$

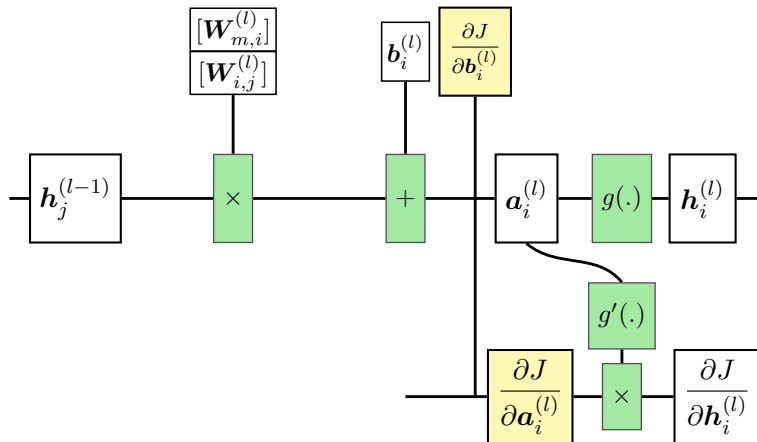
$$\Rightarrow \frac{\partial J}{\partial h_i^{(l)}} = \sum_{m=1}^{M_{l+1}} \frac{\partial J}{\partial a_m^{(l+1)}} W_{m,i}^{(l+1)} \quad \dots (4)$$

Backpropagation



$$\frac{\partial J}{\partial a_i^{(l)}} = \frac{\partial J}{\partial h_i^{(l)}} g'(\mathbf{a}_i^{(l)}) \dots (3)$$

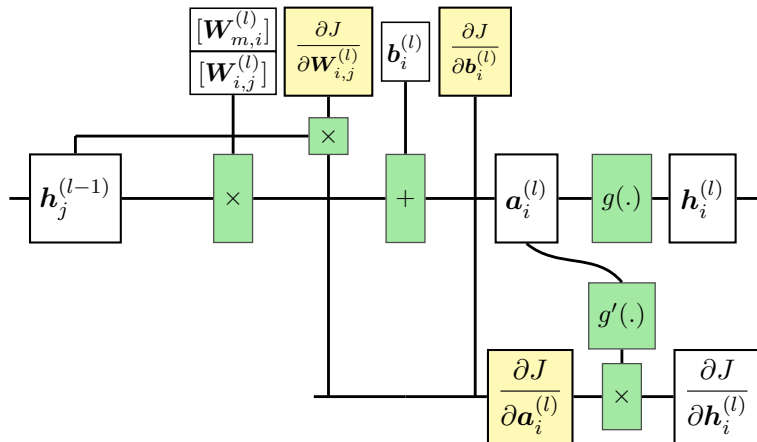
Backpropagation



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Backpropagation

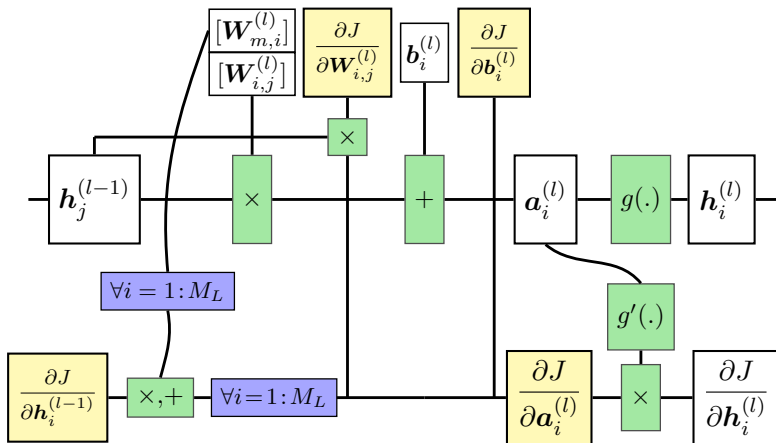


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$$\frac{\partial J}{\partial W_{i,j}^{(l)}} = \frac{\partial J}{\partial a_i^{(l)}} h_j^{(l-1)} \dots (1)$$

Backpropagation



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$$\frac{\partial J}{\partial h_i^{(l-1)}} = \sum_{m=1}^{M_l} \frac{\partial J}{\partial a_m^{(l)}} W_{m,i}^{(l)} \dots (4)$$